Example 3.68 If $A=\begin{pmatrix}2 & 1\\ 1 & 3\end{pmatrix}$, $B=\begin{pmatrix}2 & 0\\ 1 & 3\end{pmatrix}$ find AB and BA. Verify $AB=BA$?

Solution We observe that A is a 2×2 matrix and B is a 2×2 matrix, hence AB is defined and it will be of the order 2×2.

$$AB=\begin{pmatrix}2 & 1\\ 1 & 3\end{pmatrix}\times\begin{pmatrix}2 & 0\\ 1 & 3\end{pmatrix}=\begin{pmatrix}4+1 & 0+3\\ 2+3 & 0+9\end{pmatrix}=\begin{pmatrix}5 & 3\\ 5 & 9\end{pmatrix}$$

$$BA=\begin{pmatrix}2 & 0\\ 1 & 3\end{pmatrix}\times\begin{pmatrix}2 & 1\\ 1 & 3\end{pmatrix}=\begin{pmatrix}4+0 & 2+0\\ 2+3 & 1+9\end{pmatrix}=\begin{pmatrix}4 & 2\\ 5 & 10\end{pmatrix}$$

Therefore, $AB\neq BA$.

Example 3.69 If $A=\begin{pmatrix}2 & -2\sqrt{2}\\ \sqrt{2} & 2\end{pmatrix}$ and $B=\begin{pmatrix}2 & 2\sqrt{2}\\ -\sqrt{2} & 2\end{pmatrix}$

Show that A and B satisfy commutative property with respect to matrix multiplication.

Solution We have to show that $AB=BA$

$$\text{LHS}=AB=\begin{pmatrix}2 & -2\sqrt{2}\\ \sqrt{2} & 2\end{pmatrix}\times\begin{pmatrix}2 & 2\sqrt{2}\\ -\sqrt{2} & 2\end{pmatrix}=\begin{pmatrix}4+4 & 4\sqrt{2}-4\sqrt{2}\\ 2\sqrt{2}-2\sqrt{2} & 4+4\end{pmatrix}=\begin{pmatrix}8 & 0\\ 0 & 8\end{pmatrix}$$

$$\text{RHS}=BA=\begin{pmatrix}2 & 2\sqrt{2}\\ -\sqrt{2} & 2\end{pmatrix}\times\begin{pmatrix}2 & -2\sqrt{2}\\ \sqrt{2} & 2\end{pmatrix}=\begin{pmatrix}4+4 & -4\sqrt{2}+4\sqrt{2}\\ -2\sqrt{2}+2\sqrt{2} & 4+4\end{pmatrix}=\begin{pmatrix}8 & 0\\ 0 & 8\end{pmatrix}$$

Hence LHS = RHS (ie) $AB=BA$

Note

- If A and B are any two non zero matrices, then $(A+B)^2\neq A^2+2AB+B^2$.
- However if $AB=BA$ then $(A+B)^2=A^2+2AB+B^2$

Example 3.70 Solve $\begin{pmatrix}2 & 1\\ 1 & 2\end{pmatrix}\begin{pmatrix}x\\ y\end{pmatrix}=\begin{pmatrix}4\\ 5\end{pmatrix}$

Solution

$$\begin{pmatrix}2 & 1\\ 1 & 2\end{pmatrix}_{2\times2}\times\begin{pmatrix}x\\ y\end{pmatrix}_{2\times1}=\begin{pmatrix}4\\ 5\end{pmatrix}$$

By matrix multiplication $\begin{pmatrix}2x+y\\ x+2y\end{pmatrix}=\begin{pmatrix}4\\ 5\end{pmatrix}$

Rewriting $2x+y=4$ ...(1)

$x+2y=5$ ...(2)

(1) $-2\times$ (2) $\Rightarrow$ $2x+y=4$

$2x+4y=10$ (−)

$-3y=-6$ $\Rightarrow$ $y=2$

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